BILINGUAL TEXT-TO-SPEECH SYSTEM USING FASTSPEECH 2

Technical Implementation and Experimental Report

Project objective: Implement and evaluate a bilingual text-to-speech pipeline using FastSpeech 2 for English and Telugu. The system uses text representations and acoustic features to train a non-autoregressive speech synthesis model.

1. System Architecture

The pipeline consists of text processing, acoustic feature extraction, duration preparation, FastSpeech 2 training, and waveform synthesis. FastSpeech 2 predicts mel-spectrograms using an encoder, variance adaptor, and decoder. The variance adaptor models duration, pitch, and energy. A vocoder converts generated mel-spectrograms into audio.

2. Telugu Implementation

The Telugu system was trained using the IndicTTS Telugu dataset. The processed dataset contains 1,562 samples: 1,297 training samples and 265 validation samples. Audio was processed as 16 kHz mono.

The phoneme vocabulary contains 52 symbols (IDs 0–51), with ID 0 reserved for padding. Word-level timing information from MMS was used for duration preparation. Within-word phoneme durations were allocated heuristically, and inter-word gaps were assigned to the word-boundary token. This should be understood as word-level alignment with heuristic phoneme-duration allocation, not full phoneme-level forced alignment.

Pitch features were extracted using the pyworld DIO and StoneMask methods. Energy was calculated from the short-time Fourier transform. The preprocessing validation checked that duration totals matched the corresponding mel-spectrogram frame counts.

3. Telugu Training Configuration

Parameter	Configuration
Encoder	4 layers, 2 attention heads, hidden size 256
Decoder	6 layers, 2 attention heads, hidden size 256
Convolution filter	1024
Dropout	0.2
Variance predictor	Filter 256, kernel 3, dropout 0.5
Variance embedding bins	256
Batch size	8
Warm-up steps	4000
Training steps	4000

4. Telugu Training Results

The training loss decreased from approximately 19.69 to 2.16 over the training run. Validation measurements were recorded at steps 1000, 2000, 3000, and 4000.

Step	Total validation loss	Mel loss	Postnet mel loss
1000	4.1745	0.8821	0.9002
2000	4.0070	0.7942	0.8051
3000	4.1975	0.7623	0.7665

4000	4.1595	0.7773	0.7771

Step 3000 was selected for the submitted Telugu test outputs after listening-based comparison. Step 2000 had the lowest total validation loss among the reported checkpoints, while Step 3000 had the lowest mel and postnet mel validation losses. The Step 4000 checkpoint is retained as a backup.

5. Telugu Evaluation Outputs

Five official test utterances are included: T1, T2a, T2b, T3, and T4. The submission includes generated WAV files and 15 variance-adaptor plots: five duration plots, five pitch plots, and five energy plots. Training and validation loss curves are also included.

6. English Implementation and Experimental Status

The English training data was prepared from precomputed mel, pitch, energy, and duration features. Metadata was filtered to retain samples whose text-token, pitch, energy, and duration lengths were consistent and whose duration totals matched the mel frame count. This produced 1,133 training samples and 232 validation samples.

English training was run for 1,000 steps, with checkpoints saved at steps 250, 500, 750, and 1000. The recorded validation total loss was 8566.4638 at step 500 and 4622.6232 at step 1000. However, the pitch and energy losses were very large relative to the mel losses, indicating feature-scale or optimization difficulties.

The Step 500 inference output was listened to and reported as noise. The Step 1000 inference attempt did not produce a WAV file. Therefore, the English model is included as an experimental training run, not as a successfully validated speech synthesizer. No claim of natural or intelligible English synthesis is made.

7. Limitations

The English acoustic-feature losses were unstable and dominated by pitch and energy terms. English speech generation was not successful in the available experiment. Telugu duration allocation includes a heuristic within-word component. The reported results should be interpreted within these dataset and training limitations.

8. Reproducibility and Submission Contents

The project contains language-specific configuration files and training, preprocessing, and inference code directories, along with training logs, loss plots, Telugu generated audio, variance-adaptor plots, and model checkpoints. The selected Telugu Step 3000 checkpoint and Step 4000 backup are approximately 398 MB each. Large checkpoint files should be distributed through Git LFS or an external download location rather than ordinary Git commits.

The Telugu checkpoint and results are preserved in the submission workspace. English checkpoints and experiment logs are retained for transparency. Users should consult the README for environment setup and the exact paths of model weights.